

Machine Learning Project Report

Group-M

Group members:

- Lucrezia Salerno 70098
- Guilherme Morgado 56857
- Miguel Teixeira 56529
- Isaac Carvalho 57045
- Amal Bouilla 70973
- Guilherme Carvalho 70364

Table of contents:

1. Introduction.....	2
2. Business Understanding	2/3
3. Data and Methodology	3/4
4. Results and Discussion	5
5. Conclusion.....	5/6

1. Introduction

With teams spending billions of euros on player acquisitions annually, the football transfer market has become more complex and expensive. Even with the market's enormous financial scale, a sizable percentage of transactions fall short of the intended outcomes, particularly for high-value signings. These inefficiencies can hurt a club's long-term competitive success in addition to being costly.

Due to their tight budgets and inability to make needless purchases, mid-tier teams are particularly vulnerable to this problem. Because these institutions, unlike elite clubs, have to maximize the value of each transfer move, mistakes have a much greater impact.

While essential, traditional scouting techniques are usually subjective and have a limited scope. Due to their reliance on observation and experience, scouts may be biased and lose important statistical trends. Because of this, teams often overvalue players who are very well-known while ignoring others who might perform better for a lower cost.

In this context, the objective of this project is to develop a Machine Learning model capable of predicting the fair market value of professional football players. By leveraging data-driven techniques, the model aims to support more informed transfer decisions and identify players who may be undervalued in the market.

Lastly, our approach seeks to bridge the gap between decision-making and raw data, providing clubs with a more objective and scalable framework for player evaluation.

2. Business Understanding

A sports analytics system designed for mid-tier football teams with budgets between €50 million and €200 million is the setting for this project. These teams struggle greatly to compete with more affluent teams because they don't have the funds to cover expensive transfer errors.

The core problem addressed in this project is the inefficiency of player valuation in the transfer market. Clubs often rely on subjective assessments and incomplete information, which can lead to players overpaying or failing to identify valuable opportunities.

The proposed solution consists of a Machine Learning model that predicts player market value based on performance data, contextual information, and career-related features.

These include metrics such as goals, assists, minutes played, age, league quality, and contract conditions.

By comparing predicted values with actual market prices, the system enables the identification of undervalued players. These players represent opportunities for clubs to acquire talent at a lower cost while maintaining competitive performance.

From a business perspective, the success of this solution is measured by its ability to improve decision-making, reduce financial risk, and optimize resource allocation. The model effectively transforms data into actionable insights, allowing clubs to operate more strategically in the transfer market.

Another important aspect of this problem is the dynamic nature of the transfer market. Player values are influenced not only by performance but also by external factors such as league reputation, contract duration, and even market trends.

By incorporating these contextual variables into the model, the solution becomes more aligned with real-world valuation mechanisms. This allows the model to move beyond purely statistical prediction and better reflect how clubs actually evaluate players.

Therefore, the proposed approach aims to combine quantitative analysis with domain knowledge, creating a more holistic framework for player valuation.

3. Data and Methodology

The dataset used in this project is constructed by merging multiple sources of football-related data. These include player profiles, performance statistics, market value history, injury records, and team information. Each dataset contributes a different dimension of information, allowing for a more comprehensive representation of player value.

During the data loading phase, all datasets are integrated using a common player identifier. Performance data is aggregated across seasons to obtain total statistics per player, and only data from 2010 onwards is considered to ensure consistency and relevance.

Data preprocessing plays a crucial role in ensuring the quality of the dataset. This includes handling missing values, standardizing categorical variables, and converting

date fields into appropriate formats. Performance-related missing values are replaced with zeros, while categorical values are normalized to maintain consistency.

Feature engineering is a key component of the methodology. New features are created to capture important aspects of player performance and context. These include age and career phase, contract duration, league quality, and normalized performance metrics such as goals per 90 minutes and assists per 90 minutes.

In addition, interaction terms are introduced to capture more complex relationships between variables. For example, performance metrics are combined with league quality to reflect differences in competitive environments. Health-related indicators, such as injury frequency, are also included to account for player availability.

The problem is formulated as a supervised regression task, where the objective is to predict a continuous variable: player market value. Models are trained using the engineered features and evaluated using standard regression metrics such as Mean Absolute Error, Mean Squared Error, and R-squared.

This methodology ensures that the model is not only accurate but also capable of capturing real-world dynamics in player valuation.

An important aspect of the methodology is the transformation of raw features into more informative variables. For example, instead of using absolute performance metrics, the project relies on normalized indicators such as goals per 90 minutes and assists per 90 minutes. This ensures comparability across players with different playing times.

Additionally, the introduction of interaction terms allows the model to capture relationships that are not strictly linear. For instance, the impact of performance may vary depending on league quality, and combining these variables enables the model to better reflect these dynamics.

Another key consideration is data consistency. By applying systematic preprocessing steps, such as handling missing values and standardizing categorical variables, the dataset becomes more reliable and suitable for modeling.

4. Results and Discussion

The results obtained from the modeling process indicate that the model is capable of capturing meaningful relationships between player attributes and market value. Feature engineering plays a significant role in improving predictive performance, particularly through the use of normalized performance metrics and contextual variables such as league quality.

Age and career stage are also strong predictors of player value, reflecting real-world patterns in football. Younger players often have higher potential value, while players in their prime tend to command the highest market prices.

By comparing predicted values with actual market prices, the model identifies discrepancies that indicate potential undervaluation or overvaluation. This provides valuable insights for decision-making, as clubs can focus on players whose predicted value exceeds their current market price.

However, it is important to note that model performance may vary depending on the choice of algorithm and parameter tuning. While the current approach demonstrates strong potential, further optimization could improve accuracy and robustness.

Overall, the results highlight the importance of combining data-driven methods with domain knowledge to achieve meaningful insights. The model not only predicts values but also provides a framework for understanding the factors that influence player valuation.

5. Conclusion

This project demonstrates the application of Machine Learning to a real-world problem in sports analytics. By combining multiple data sources, structured preprocessing, and advanced feature engineering, a predictive system for player valuation was developed.

The key contribution of this project is not only accurate prediction, but also the ability to identify inefficiencies in the transfer market. By highlighting undervalued players, the model enables clubs to make more informed and strategic decisions.

From a technical perspective, the project showcases a complete Machine Learning pipeline, including data integration, preprocessing, feature engineering, modeling, and

evaluation. This reflects best practices in the field and provides a strong foundation for future development.

Future work may include incorporating more advanced performance metrics, integrating real-time data, and applying more sophisticated modeling techniques. These improvements could further enhance the accuracy and applicability of the system.

In conclusion, this project demonstrates the potential of Machine Learning to transform football analytics by turning data into actionable insights. It provides a scalable and practical solution for improving player valuation and supporting smarter recruitment strategies.